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QP-014 Risk Management (Strategic and Operational)

Standards and clause references are guidance only and do not claim certification or standard compliance. Applicable customer, contractual and legal requirements govern accepted work.

1. Purpose

Define the process for identifying, assessing, mitigating, and monitoring risks at both the strategic/enterprise level (AS9100D Clause 6.1) and the operational/job level (AS9100D Clause 8.1.1) to ensure product quality, on-time delivery, and business continuity.

2. Scope

Applies to:

- **Part A — Strategic Risks (Clause 6.1):** Organization-wide risks and opportunities affecting the QMS's ability to achieve its intended results.
- **Part B — Operational Risks (Clause 8.1.1):** Process-specific and job-specific risks that could affect product conformity, delivery performance, or customer satisfaction.

3. Responsibilities

Role	Responsibility
Shop Owner	Identify risks, perform assessments, implement mitigations, review risk register quarterly, update after triggering events

4. References

- QF-023 - Risk Register
- Risk_Matrix_5x5 (04-supporting/)
- QP-012 - Management Review
- QP-009 - Corrective Action
- AS9100 Rev D Clause 6.1 (Actions to Address Risks and Opportunities)
- AS9100 Rev D Clause 8.1.1 (Operational Risk Management)

5. Definitions

- **Risk:** The effect of uncertainty on objectives, expressed as a combination of likelihood of occurrence and severity of consequence.
- **Likelihood:** The probability that a risk event will occur (rated 1-5).
- **Severity:** The magnitude of impact if the risk event occurs (rated 1-5).
- **Risk Priority Number (RPN):** Likelihood x Severity (range 1-25). Used to prioritize mitigation efforts.
- **Mitigation:** Actions taken to reduce either the likelihood or severity of a risk to an acceptable level.

6. Procedure

PART A — Strategic Risk Management (Clause 6.1)

6.1 Strategic Risk Identification

1. Identify operational risks by considering:
    - Process risks (machine failure, programming errors, setup errors)
    - Supply chain risks (single-source suppliers, material availability, long lead times)
    - Equipment risks (calibration drift, machine breakdown, tooling failure)
    - Personnel risks (single-operator illness/injury, skill gaps)
    - Customer risks (requirement changes, unclear specifications, payment)
    - External risks (regulatory changes, material cost increases, utility disruption)
2. Document each identified risk on QF-023 Risk Register with: risk ID, description, category, affected process/objective.

6.2 Strategic Risk Assessment

1. For each identified risk, assess using the 5x5 matrix (reference Risk_Matrix_5x5 in 04-supporting/):
    - **Likelihood** (1=Rare, 2=Unlikely, 3=Possible, 4=Likely, 5=Almost Certain)
    - **Severity** (1=Negligible, 2=Minor, 3=Moderate, 4=Major, 5=Critical)
2. Calculate Risk Priority Number: RPN = Likelihood x Severity.
3. Categorize risk level:
    - Low (RPN 1-4): Accept, monitor at quarterly review
    - Medium (RPN 5-9): Implement mitigation within 60 days
    - High (RPN 10-15): Implement mitigation within 30 days
    - Critical (RPN 16-25): Immediate action required, do not proceed with affected process until mitigated

6.3 Strategic Risk Mitigation

1. For each risk requiring mitigation (Medium and above), define:
    - Mitigation action(s) to reduce likelihood, severity, or both
    - Target completion date
    - Responsible person (Shop Owner)
2. Common mitigation strategies:
    - Elimination: Remove the source of risk
    - Substitution: Replace with lower-risk alternative
    - Engineering controls: Add process controls, fixtures, verification steps
    - Administrative controls: Procedures, training, checklists
    - Contingency planning: Define response if risk event occurs
3. Record mitigation actions on QF-023 Risk Register.

6.4 Strategic Risk Mitigation Implementation and Verification

1. Implement planned mitigations by target date.
2. After implementation, re-assess the residual risk (new Likelihood x Severity) and record on QF-023.
3. If residual risk remains Medium or above, define additional mitigations or formally accept the residual risk with documented justification.

6.5 Review and Monitoring

1. Review the full Strategic Risk Register quarterly at minimum.
2. Conduct ad-hoc risk assessments when any of the following occur:
    - New customer or new product type
    - New material or process introduced
    - New or changed supplier
    - Nonconformance trend identified (3 or more related NCRs)

    - Significant equipment change or addition
    - External event affecting operations
3. Update risk ratings based on current data (nonconformance trends, delivery performance, supplier performance).
4. Report risk status as a required input at Management Review (QP-012).

PART B — Operational Risk Management (Clause 8.1.1)

6.6 Job-Level Operational Risk Assessment

1. When a new purchase order or job is received, perform an operational risk assessment considering:
    - Is this a new part number or repeat production?
    - Does the job involve material the shop has not previously processed?
    - Are there tight tolerances at or near machine capability limits?
    - Is the delivery timeline compressed, requiring overtime or expedited material?
    - Are special processes required from unfamiliar suppliers?
    - Does the drawing revision differ from the last production run?
    - Are there customer-identified critical/safety characteristics?
2. Document the job risk assessment on the Job Traveler (QF-004) risk field or on a separate job risk assessment note attached to the traveler.
3. For jobs assessed as elevated risk, define additional controls:
    - Additional verification of high-risk characteristics; full inspection remains the default unless the customer specifies or approves a different plan
    - First-piece verification hold points
    - Additional setup verification steps
    - A separate, deliberate owner review of the CNC program before prove-out
    - Contingency material procurement

6.7 Operational Risk Criteria and Acceptance

1. Define acceptable and unacceptable operational risk levels:
    - **Acceptable:** Repeat job, proven material, tolerances within demonstrated capability, standard delivery timeline
    - **Elevated:** New part, new material, tight tolerances, compressed timeline — proceed with additional controls per Step 18
    - **Unacceptable:** Tolerances beyond machine capability, material shop cannot source from approved suppliers, process requirements shop cannot meet — decline the order or negotiate requirements with customer before acceptance
2. Formally accept residual operational risk for each elevated-risk job by sign-off on the traveler.

6.8 Communication of Operational Risks

1. Communicate identified operational risks that could affect product conformity to relevant parties:
    - To customer: when risk assessment reveals potential inability to meet requirements
    - To supplier: when incoming material or process quality is critical to risk mitigation
2. Record risk communications on the Job Traveler or in customer/supplier communication logs.

6.9 Records

1. Maintain QF-023 Risk Register as a living document, updated at each review (Part A).
2. Maintain job-level risk assessments on travelers or attached notes (Part B).
3. Retain superseded risk assessments and register versions under QP-001 and SUP-003 for at least ten years after completion or the relevant last shipment, applying the longest applicable requirement.

7. Records Generated

Form No.	Title	When Completed
QF-023	Risk Register	Initial assessment and each quarterly review

8. Revision History

Rev	Date	Description	Approved By
A	2026-08-07	Initial release	Shop Owner

Revision B Approval

Approved 2026-11-05 by Jacob Honick, Owner. Standardize control metadata and annual review, remove fixed page counts and align applicable references and wording.